

PILOT-GM-VAE: Patient-Level Analysis of single cell Disease Atlas with Optimal Transport of Gaussian Mixture Variational Autoencoders

1 Data sets

We use public single-cell data sets to benchmark the proposed methods.

Peng and colleagues (Peng et al., 2019) performed a single-cell study on pancreatic ductal adenocarcinoma (PDAC). They characterized 11 healthy and 24 PDAC samples with a total of 57,530 scRNA-seq, which were clustered and annotated in 10 major pancreas cell clusters. As only raw data was provided (obtained from GSA: CRA001160 <https://ngdc.cncb.ac.cn/gsa/browse/CRA001160>), we reanalyzed the data using Seurat(Stuart et al., 2019) (filtering cells and normalizing the data) and re-annotated cell clusters using the same marker genes.

Systemic lupus erythematosus is a common type of lupus that, in the immune system, pounds its tissues, yielding overall rash and tissue injury in the acted organs (Perez et al., 2022). This single-cell study characterized peripheral blood mononuclear cells (PBMCs) gene expression of 261 donors consisting of 1,263,676 cells. Of these donors, 99 were healthy controls, and 162 were disease patients. The pre-processing of the data, including the filtering of cells and normalization, was done by (Wolf et al., 2018). We use the normalized and clustered (11 groups) data sets provided in Gene Expression Omics (GEO; GSE174188; GSE174188_C*LUES1_a**adjusted.h5ad.gz*).

We used single cell and single-nucleus assays from kidney samples provided by the Kidney Precision Medicine Project (Lake et al., 2023). Specifically, we included single cell RNA-seq from kidney tissues encompassing 104,314 cells (Seurat(Stuart et al., 2019) for filtering cells and normalizing the data), organized into 14 major cell types. This dataset consists of samples from 45 donors: 18 controls, 12 with acute kidney failure, and 15 with chronic kidney disease. This data were obtained from CellxGene.

The lung cancer single-cell atlas is another large data set with 941,504 cells from 165 donors (Sikkema et al., 2023). The samples are formed by 51 normal lung, 18 chronic obstructive pulmonary disease, 76 lung adenocarcinomas, 13 non-small cell lung carcinomas, and 12 squamous cell lung carcinoma. The cells are clustered into 33 clusters/cell types. The atlas is based on distinct studies measured under distinct platforms. We only consider here samples from lung tissue and measured with the 10X genomics platform. Data was obtained from Zenodo (ID 7227571). The pre-processing of the data, including the filtering of cells and normalization, was done by (Wolf et al., 2018).

We have recently proposed a large study to characterize myocardial infarction (Kuppe et al., 2022). We recovered 115,517 high-quality cells, which were clustered in 33 cellular clusters. We focus here on samples associated with healthy heart tissues (healthy controls and remote zones; n=13) and ischemic zone (IZ; n=7). We exclude one sample due to low quality (less than 1000 genes (on average) per cell). The pre-processing of the data, including the filtering of cells and normalization, was done by (Wolf et al., 2018). A final data set is provided in Zenodo (ID 7435911).

We include an extended version of the myocardial infarction atlas((Kuppe et al., 2022), which includes 23 samples representing controls (13), ischemic (5), and fibrotic (5) samples (MI-2). The data contains 132,888 cells clustered into 11 cell types. The pre-processing of the data, including the filtering of cells and normalization, was done by (Wolf et al., 2018). The data was obtained from CellxGene.

The single-cell pancreas diabetes data (Hrovatin et al., 2023) covers 264,235 mouse pancreatic islet cells (with 13 cell clusters) from 52 samples. This includes 12 endocrine pancreas disorders, 6 type-1 diabetes mellitus, 12 type-2 diabetes mellitus, and 22 normal samples. We excluded four samples (Embryos E12-E15) due to being outliers by the trajectories of the methods. The data set was obtained in GEO (GSE211799). Follicular lymphoma data (Han et al, 2022) collected by Han and colleagues, containing 137,147 single cells have been processed to illustrate the various tumor and immune cell populations of Follicular lymphoma. Cells have been clustered into 18 cell clusters from 23 donors (3 normal and 20 lymphoma samples). The pre-processing of the data, including the filtering of cells and normalization, was done by (Wolf et al., 2018). The raw data is accessible from EGA (EGAS00001006052).

The single cell transcriptional profiling of clear cell renal cell carcinoma (ccRCC) (Zvirblyte et al., 2024) provides a comprehensive dataset characterizing the tumor microenvironment using single cell RNA sequencing. The dataset encompasses 17 samples, including eight normal and nine tumor samples, and identifies six major cell types. The pre-processing of the data, including the filtering of cells and normalization, was done by (Wolf et al., 2018). The pre-processed data were obtained from GEO (GSE242299).

The breast cancer atlas provides a detailed genomic and spatial characterization of the adult human breast. This comprehensive dataset comprises 714,331 single cells from 126 donors (Kumar et al., 2023), including samples from 86 normal breast tissues and 40 cancerous tissues. The cells are categorized into 10 distinct cell types. The pre-processing of the data, including the filtering of cells and normalization, was done by Seurat (Stuart et al., 2019). The pre-processed data were obtained from GEO (GSE195665).

We obtained a study with PBMC cells from patients infected with Covid-19 (Ren et al., 2021). We only considered patients with PMBCs (frozen or fresh cells). This totals 151 samples classified as either severe infection (n=70), mild infection (n=61), and control (n=20). For these samples, we have 993,171 cells, which were grouped into 10 major cell clusters. The pre-processing of the data, including the filtering of cells and normalization, was done by (Wolf et al., 2018). The data were obtained from GEO (GSE158055).

The Alzheimer’s disease atlas offers a comprehensive multimodal characterization of the human brain, integrating single-cell and spatial transcriptomic data (Gabbito et al., 2024). This dataset comprises 1,395,601 single nuclei from control samples (n=44) and Alzheimer’s disease patients (n=39). The cells are annotated into 18 cell types. The pre-processing of the data, including the filtering of cells and normalization, was done by (Stuart et al., 2019). Processed data were obtained from CellxGene.

The Parkinson’s disease transcriptomic resource provides a regionally resolved single-nucleus RNA sequencing profile of the human brain (NM et al., 2024). This dataset includes 2,096,155 nuclei derived from 97 donors, comprising 73 individuals with Parkinson’s disease pathology and 24 control subjects. Profiling was performed across five anatomically distinct brain regions to capture both early and advanced disease stages. The dataset enables the analysis of gene expression across 11 identified cell types. The pre-processing of the data, including the filtering of cells and normalization, was done by (Wolf et al., 2018). Pre-processed data were obtained from CellxGene.

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